Process Priority Classifier

11-Feature ONNX Neural Network

Snapdragon AI Smart Process & Task Manager
Model Architecture & Validation Report

1. Model Overview

The Process Priority Classifier is a lightweight neural network designed to classify Windows processes into four workload categories based on 11 observable system features. The model runs entirely on-device via ONNX Runtime, with a target execution path on the Qualcomm Snapdragon Hexagon NPU.

Model size is 1.9 KB, making it suitable for edge deployment on resource-constrained Windows on Arm devices. Training uses pure NumPy (no scikit-learn or PyTorch dependencies), and the exported ONNX graph contains only four operators: Gemm, Relu, Softmax, and ArgMax.

2. Network Architecture

The model is a 3-layer MLP with the following structure:

Layer	Type	Input	Output	Activation
1	Gemm + ReLU	11	16	ReLU
2	Gemm + ReLU	16	8	ReLU
3	Gemm + Softmax	8	4	Softmax
4	ArgMax	4	1	Index

Total parameters: 11x16 + 16 + 16x8 + 8 + 8x4 + 4 = 176 + 16 + 128 + 8 + 32 + 4 = 364. ONNX opset 12, IR version 10, model size 1.9 KB.

3. Input Features (11 signals)

Each process is represented by an 11-dimensional feature vector extracted from live system telemetry. Features are normalized using pre-computed mean and scale values baked into the model metadata.

#	Feature	Description
1	cpu_norm	Normalized CPU usage (% of total capacity)
2	cpu_trend	Short-term CPU trend (rising / falling)
3	mem_norm	Normalized memory usage
4	io_rate	Disk I/O rate level
5	net_level	Network activity level
6	is_foreground	Whether the process owns the foreground window
7	on_battery	Whether the system is on battery power
8	user_activity	Recent user input activity indicator
9	arch_native	Whether the binary runs natively (not emulated)
10	is_emulated	Whether the process is running under x86/x64 emulation
11	hist_cpu	Historical CPU average for this process

4. Output Classes

Class	Label	Typical Meaning
0	PROTECTED	System-critical or foreground application
1	USER_IMPORTANT	Known important background workload
2	FLEXIBLE	Safe to reduce background activity
3	DEFERRABLE	Can be deferred when appropriate

5. Training Details

Framework: Pure NumPy implementation (no external ML frameworks)

Optimizer: Adam (learning rate 0.001)

Epochs: 250 with batch size 64

Loss: Cross-entropy with temperature scaling

Dataset: 4,000 synthetic samples (1,000 per class)

Synthetic test accuracy: 87.38% (explicitly NOT real-world accuracy)

No label leakage in final training pipeline

Temperature calibration: T = 2.0 (NLL grid search)

OOD centroids computed per-class in standardized feature space

6. Confidence Calibration & OOD Detection

Raw softmax probabilities are never trusted alone. The final confidence score is computed as a product of calibrated model confidence and familiarity:

```
final_confidence = calibrated_confidence x familiarity
```

Familiarity is computed using a radial basis function (RBF) kernel measuring distance to the nearest per-class centroid in standardized feature space. Unfamiliar workloads (low familiarity) receive reduced confidence, causing the system to become more conservative.

Parameter	Value	Description
Temperature (T)	2.0	Softmax temperature scaling
OOD threshold	0.35	Minimum familiarity for known classification
Confidence threshold	0.85	Minimum for optimization candidate
Extreme z-score	|z| > 5	Halves familiarity score

7. Safety Layer (Post-Model)

The AI model only predicts. A separate deterministic safety engine makes the final decision. The safety layer operates on a default-deny posture:

Kernel pseudo-processes (PID 0, 4): always PROTECT, never sent to model

Allowlisted system processes (lsass, csrss, services, etc.): always PROTECT

Foreground applications: always PROTECT (no action on what the user sees)

Out-of-distribution workloads: always REVIEW (never act on the unfamiliar)

Low confidence (< 0.85): always REVIEW

High confidence + high familiarity + safe class: OPTIMIZATION CANDIDATE

No process-control operations exist anywhere in the codebase. The system is recommendation-only and non-destructive.

8. Qualcomm Snapdragon NPU Validation

The ONNX model was validated through Qualcomm AI Hub on a Snapdragon X Elite CRD (sc8380xp, Windows 11, Hexagon v73). This is model-level validation on reference hardware, not a claim about local execution.

Stage	Job ID	Status	Key Result
Compile	jp0m9yd0g	SUCCESS	ONNX model compiled to DLC
Link	j568m6rng	SUCCESS	Target DLC: 13.1 KB
Profile	jpxlrj1lp	SUCCESS	10/10 nodes on NPU
Inference	j5wlrew3p	SUCCESS	4/4 argmax agreement

Maximum probability difference between CPU and NPU outputs: 0.0015. Early compilation failures (int64 flag, batch shape) are retained in job history as documented debugging evidence.

9. Performance Characteristics (Local Intel CPU)

Metric	Value
Model size	1.9 KB
ONNX operators	4 (Gemm, Relu, Softmax, ArgMax)
ONNX nodes	10
Opset / IR version	12 / 10
Single-row CPU inference	0.024 ms
Batch-64 CPU inference	0.051 ms
Model load time	13.06 ms
Monitoring overhead	~1% steady CPU
Application RAM usage	~70 MB
Total parameters	364

10. Deployment Architecture

The model is deployed as part of a local-only HTTP server application. The architecture separates telemetry collection, feature engineering, ONNX inference, safety decisions, and recommendation generation into distinct modules.

Pipeline flow:

```
psutil telemetry  ->  11-feature vector  ->  ONNX inference  ->  softmax + ArgMax
                                                    |
                                      calibrated confidence x familiarity
                                                    |
                                          deterministic safety layer
                                                    |
                                          read-only recommendation
```

11. Honest Limitations

Model trained on synthetic data only (87.38% synthetic fit, not real-world accuracy)

Full HP Snapdragon PC end-to-end validation requires physical hardware

No battery savings or CPU savings measurements have been performed

GPU telemetry not available via psutil on all Windows systems

The local development machine is Intel-based; CPU FALLBACK is the honest status

The 11-feature model is intentionally simple; real-world accuracy is unmeasured

12. Model Artifacts

File	Description
models/process_classifier.onnx	The 1.9 KB ONNX model (opset 12, IR v10)
models/model_metadata.json	Normalization stats, OOD centroids, calibration
ai/train_model.py	Pure NumPy training + ONNX graph export script
ai/dataset.py	Synthetic dataset generator (4 classes x 1000 samples)
ai/features.py	11-feature engineering module
ai/ood.py	Out-of-distribution detection (RBF centroid distance)
ai/safety.py	Deterministic safety layer (4 states)
ai/recommend.py	Recommendation mapping with confidence threshold
inference/onnx_backend.py	ONNX Runtime inference (CPUExecutionProvider)
inference/backend.py	Backend selection (NPU vs CPU fallback)

Independent competition prototype. Not affiliated with or endorsed by Qualcomm, Snapdragon, HP, or Microsoft. All validation evidence is from Qualcomm AI Hub reference hardware (Snapdragon X Elite CRD), not local execution.